

Econ 101A

Section 5

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1 Multivariate Optimization

- Multivariate optimization is not conceptually different than univariate optimization, but we do need a few more tools in order to solve multivariate problems.
- Consider $y = f(\mathbf{x}; \boldsymbol{\alpha}) = f(x_1, x_2, \dots, x_m; \alpha_1, \alpha_2, \dots, \alpha_k)$, where the x 's are variables and the α 's are parameters, aka constants.
- Function f reaches its maximum (minimum) at an interior point x^* if:
 - FOC: $\nabla f|_{x=x^*} = \mathbf{0}$
 - SOC: $H_f|_{x=x^*}$ is negative (positive) definite for a maximum (minimum).

- Important: Your main responsibility in the class is to know the case for a function with two independent variables. In this case, the FOC and SOC boil down to the conditions we discussed in lecture. That is, a function $f(x_1, x_2)$ reaches a local maximum (minimum) at (x_1^*, x_2^*) if:

- FOC: $f'_{x_1}(x_1^*, x_2^*) = 0$ and $f'_{x_2}(x_1^*, x_2^*) = 0$
- SOC: $f''_{x_1, x_1}(x_1^*, x_2^*) < (>) 0$ and $f''_{x_1, x_1}(x_1^*, x_2^*)f''_{x_2, x_2}(x_1^*, x_2^*) - (f''_{x_1, x_2}(x_1^*, x_2^*))^2 > 0$

- Definition:
 - H is a $n \times n$ square matrix.
 - H is positive definite if $x'Hx > 0$ for any $n \times 1$ non-zero vector x . H is positive semidefinite if $x'Hx \geq 0$ for any $n \times 1$ non-zero vector x .
 - H is negative definite if $x'Hx < 0$ for any $n \times 1$ non-zero vector x . H is negative semidefinite if $x'Hx \leq 0$ for any $n \times 1$ non-zero vector x .

- (Optional) Here are some rules about matrices that we can apply to the Hessian and thus determine whether the SOC is met...

Consider a symmetric matrix: $H = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

- A principal submatrix is a matrix that comes from deleting some of the rows and the same corresponding columns of that matrix (e.g. delete row 2 and column 2 or delete columns 1 and 3 and rows 1 and 3).
- A leading principal submatrix is one that comes from deleting some of the last rows and columns (e.g. delete row 3 and column 3 or delete rows 2 and 3 and columns 2 and 3).

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- A principal minor is the determinant of a principal submatrix and a leading principal minor is the determinant of a leading principal submatrix.
- The k^{th} order leading principal submatrix comes from deleting all but the first k rows and columns, so the k^{th} order leading principal minors of H are:

$$* k = 1: |H|_1 = |a| = a$$

$$* k = 2: |H|_2 = \begin{vmatrix} a & b \\ d & e \end{vmatrix} = a \cdot e - b \cdot d$$

$$* k = 3: |H|_3 = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \cdot \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \cdot \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \cdot \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

Importantly:

- A symmetric matrix is positive-definite iff leading principal minors are all positive - i.e. $|H|_1 > 0, |H|_2 > 0, |H|_3 > 0$. (This is called Sylvester's Criterion.)
- A symmetric matrix is negative-definite iff principal minors of order k have sign $(-1)^k$ - i.e. $|H|_1 < 0, |H|_2 > 0, |H|_3 < 0$.
- As in the univariate case:
 - a twice differentiable function is concave (convex) if and only if its hessian is semidefinite negative (positive),
 - the FOC and concavity (convexity) insure global optimality.

2 Constrained Maximization with Comparative Statics

Suppose that you want to maximize $f(x; p)$ such that $h(x; p) = 0$, and x is an n -dimensional vector.¹ The steps:

1. Write the Lagrangian function:

$$\mathcal{L}(\lambda, x; p) = f(x; p) - \lambda h(x; p).$$

Note that λ is joining your x 's as an additional endogenous variable.

2. Your (necessary) first order conditions (setting $\nabla \mathcal{L}(\lambda^*, x^*; p)$) will now give your $n + 1$ equations in $n + 1$ unknowns. To get your candidate solutions, take the first derivatives of $\mathcal{L}$ with respect to all x_i 's and λ and set all $n + 1$ of these equations equal to zero. You will notice that you will get the following:

$$\begin{aligned} h(x^*; p) &= 0 \\ \frac{\partial f(x^*; p)}{\partial x_i} - \lambda^* \times \frac{\partial h(x^*; p)}{\partial x_i} &= 0 \quad \text{for all } i \end{aligned}$$

3. To check the sufficient second order condition for a constrained maximum (or minimum), we use the bordered Hessian, which is calculated simply as the Hessian of $\mathcal{L}(\lambda, x; p)$. The determinant of the bordered Hessian is positive for a constrained maximum.
4. Now we want to calculate $dx_i^*(p)/dp$. We need to apply multivariate version of Implicit Function Theorem because we have more than 1 endogenous variables (for a general version, see Appendix).

The intuition behind IFT still applies. First, find a set of **equations** (FOC):

$$\underbrace{\mathbf{H}(x, p)}_{(n+1) \times 1} = \begin{cases} h(x; p) \\ \partial f(x; p)/\partial x_1 - \lambda \cdot \partial h(x; p)/\partial x_1 \\ \dots \\ \partial f(x; p)/\partial x_n - \lambda \cdot \partial h(x; p)/\partial x_n \end{cases} = \underbrace{\mathbf{0}}_{(n+1) \times 1}$$

¹Here, we will assume that h is scalar-valued, meaning that we only have one constraint. In lecture, we were more general, and allowed ourselves to have more than one constraint.

Then, if you change the value p , it would have a direct effect $D_p \mathbf{H}(x, p)$, and an indirect effect through all x_i (p can first affects any x_i , which we have $n + 1$ endogenous variables, and then each of the $n + 1$ endogenous variables can affect each of the $n + 1$ equations in $\mathbf{H}$). Two effects cancel out eventually because we're operating on a set of equations:

$$\underbrace{D_x \mathbf{H}(x, p)}_{(n+1) \times (n+1)} \underbrace{D_p x}_{(n+1) \times 1} = - \underbrace{D_p \mathbf{H}(x, p)}_{(n+1) \times 1}$$

Now this is a classic system of equations we see in linear algebra. We can use Cramer's Rule (see Appendix) to solve for $dx_i^*(p)/dp$.

- Now we want to calculate $df(x^*(p); p)/dp$. Envelope Theorem applies to the partial derivative of the Lagrangian, NOT the partial in the objective function. In other words, the Envelope Theorem for constrained optimization is as follows:

$$\frac{df(x^*(p); p)}{dp} = \frac{\partial \mathcal{L}(\lambda^*(p), x^*(p); p)}{\partial p}$$

3 Exercises

3.1 Problem 1 (Problem 2 from PS1, Spring 2014)

Let a and b be two real non null parameters. Consider the following maximization problem: $\max_{x,y} f(x, y; a, b) = ax^2 - x + by^2 - y$

- Write down the first order conditions for this problem with respect to x and y (notice that a and b are parameters).
- Solve explicitly for the x^* and the y^* that satisfy the first order conditions.
- Now the second order conditions at (x^*, y^*) . Give sufficient conditions on a and b for the stationary point that you found in point 2 to be a maximum?
- Under what conditions on a and b is the function f concave in x and y ? When is it convex in x and y ?

3.2 Problem 2

Consider the function f defined on $[-1, 1] \times [-1, 1]$ by: $f(x, y) = x^2 - y^2$.

- Provide the maximal points of f and the points (x, y) at which the maximum is reach (no calculus needed).
- Let's now try to use our recipe:
 - Write down the first order conditions for this problem with respect to x and y and solve for the x^* and the y^* that satisfy the first order conditions.
 - Compute the hessian of f at the stationary points you found. Is the hessian positive definite? Is it negative definite? What can you infer?

3.3 Problem 3

Suppose you are taking only two classes: literature and economics. It is finals week, and you know you can study productively for T hours a day. You must choose how to split these hours between studying for literature (which involves a lot of reading) and studying economics. Your goal is to maximize your GPA. An exogenous variable, the amount of free coffee you get, helps you read more efficiently but it

does not affect your studying of economics. Assume that you know exactly how your daily studying of each subject will determine your GPA this semester:

$$GPA = \frac{2}{3}\sqrt{C}\sqrt{h_L} + \frac{2}{3}\sqrt{h_E},$$

where h_L = hours studying literature, h_E = hours studying economics, and C = cups of coffee.

1. What would be the Lagrangian for this maximization problem? Find the potential optimum and check the second order condition.
2. How will the hours spent studying literature change if the exogenous amount of coffee increases? Use the IFT to answer this question (you'll either need to use Cramer's rule or to inverse a 3 by 3 matrix). Can you solve for the explicit form of h_L^* as a function of C to verify this answer?
3. How will your maximized GPA change if the exogenous amount of coffee C increases? What if the number of total productive hours T increases? Use the Envelope Theorem (the constrained version) to answer this question.

4 Appendix

4.1 Multivariate IFT

Assume we have a function $\mathbf{f} : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^n$, and $\mathbf{f}(\mathbf{x}; \mathbf{p}) = \mathbf{0}_n$.² We can breakdown

$$\mathbf{f} = \begin{pmatrix} f_1(x_1, \dots, x_n; p_1, \dots, p_m) \\ f_2(x_1, \dots, x_n; p_1, \dots, p_m) \\ \dots \\ f_n(x_1, \dots, x_n; p_1, \dots, p_m) \end{pmatrix}$$

- If the following 2 conditions hold:
 1. All f_i for $i = 1, \dots, n$ are continuously differentiable around $(\mathbf{x}_0, \mathbf{p}_0)$ (a solution to $\mathbf{f}(\mathbf{x}; \mathbf{p}) = \mathbf{0}$).
 2. $D_{\mathbf{x}}\mathbf{f}(\mathbf{x}_0; \mathbf{p}_0)$, the Jacobian matrix evaluated at $(\mathbf{x}_0, \mathbf{p}_0)$, has full rank (alternatively, the determinant of the Jacobian Matrix is non-zero or the determinant of the Jacobian matrix is invertible)
- Then we have 2 useful results:
 1. There exists $\mathbf{g}(\mathbf{p})$ such that $\mathbf{x} = \mathbf{g}(\mathbf{p})$ in some neighborhood around $\mathbf{p}_0$.
 2. All the derivatives you could ever want are summarized as:

$$D_{\mathbf{p}}\mathbf{g}(\mathbf{p}_0) = -[D_{\mathbf{x}}\mathbf{f}(\mathbf{x}_0; \mathbf{p}_0)]^{-1} D_{\mathbf{p}}\mathbf{f}(\mathbf{x}_0, \mathbf{p}_0) \quad (*)$$

If this doesn't make immediate sense to you. Think about all of your n x 's (e.g. demand equations). Suppose you want to find out each of these changes with respect to the third parameter (e.g. price of good x_3). Then you're solving for the third column of $D_{\mathbf{p}}\mathbf{g}(\mathbf{p}_0)$ by using the third column of $D_{\mathbf{p}}\mathbf{f}(\mathbf{x}_0; \mathbf{p}_0)$. Writing this out explicitly:

$$\begin{pmatrix} \frac{\partial g_1}{\partial p_3} \\ \frac{\partial g_2}{\partial p_3} \\ \dots \\ \frac{\partial g_n}{\partial p_3} \end{pmatrix} = -[D_{\mathbf{x}}\mathbf{f}(\mathbf{x}_0; \mathbf{p}_0)]^{-1} \begin{pmatrix} \frac{\partial f_1}{\partial p_3} \\ \frac{\partial f_2}{\partial p_3} \\ \dots \\ \frac{\partial f_n}{\partial p_3} \end{pmatrix}$$

We just do this for every single choice and every single parameter which is why we have an n (goods) $\times$ m (parameter) matrix. Using Cramer's Rule, we solve for $x_i = g_i(\mathbf{p})$, the partial derivative of x_i with respect to p_k (see below).

²In class, Professor Gaubert writes $\frac{\partial \mathbf{f}}{\partial \mathbf{x}}$ to write the Jacobian, which we have been writing as J_f or $D_{\mathbf{x}}\mathbf{f}(\mathbf{x}_0; \mathbf{p}_0)$. These are all the same thing, and it's good to get used to new notation. Also $\mathbf{0}_n$ represents a zero vector of size n .

4.2 Cramer's Rule

Suppose you have a system of equations $\mathbf{A}\mathbf{X} = \mathbf{B}$. Then

$$x_{ij} = -\frac{|\mathbf{A}_{ij}|}{|\mathbf{A}|}$$

where $\mathbf{A}_{ij}$ is the matrix that comes from replacing the i^{th} column of $\mathbf{A}$ with the j^{th} column of $\mathbf{B}$.

How do we use this? Well let's start relabing things from the multivariate implicit function theorem:

$$\underbrace{D_{\mathbf{p}}\mathbf{g}(\mathbf{p}_0)}_{\mathbf{X}} = \underbrace{[D_{\mathbf{x}}\mathbf{f}(\mathbf{x}_0; \mathbf{p}_0)]^{-1}}_{\mathbf{A}} \underbrace{D_{\mathbf{p}}\mathbf{f}(\mathbf{x}_0, \mathbf{p}_0)}_{\mathbf{B}}$$

I just rewrote what I had before: as $\mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$, or $\mathbf{A}\mathbf{X} = \mathbf{B}$. Using Cramer's Rule, we can reproduce the formula we saw in class since

$$\mathbf{A}_{ij} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_{i-1}} & \frac{\partial f_1}{\partial p_j} & \frac{\partial f_1}{\partial x_{i+1}} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_{i-1}} & \frac{\partial f_n}{\partial p_j} & \frac{\partial f_n}{\partial x_{i+1}} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

Just to reproduce the formula we saw in class:

$$\frac{\partial g_i}{\partial p_k} = -\frac{\begin{vmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_{i-1}} & \frac{\partial f_1}{\partial p_j} & \frac{\partial f_1}{\partial x_{i+1}} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_{i-1}} & \frac{\partial f_n}{\partial p_j} & \frac{\partial f_n}{\partial x_{i+1}} & \cdots & \frac{\partial f_n}{\partial x_n} \end{vmatrix}}{|D_{\mathbf{x}}\mathbf{f}(\mathbf{x}_0; \mathbf{p}_0)|}$$